CS 148a PS4

Om Sanan

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§1 Data Annotation

(a)

(i) The Intersection Over Union (IoU) metric evaluates how well a model's predicted bounding box aligns with the actual bounding box of an object in an input image. In the image below, the predicted box is shown in red, and the ground truth box is in green. Calculate the IoU of these two bounding boxes and provide a numerical answer.

Both boxes have area

$$10 \times 10 = 100$$

The overlapping region goes from $(5, 5)$ to $(10, 10)$ so intersection area is:

$$5 \times 5 = 25$$

The union area is:

$$100 + 100 - 25 = 175$$

Therefore, the Intersection over Union is:

$$\text{IoU} = \frac{\text{Intersection}}{\text{Union}} = \frac{25}{175} = \frac{1}{7} \approx 0.143$$

$$\boxed{\text{IoU} \approx 0.143}$$

(ii) What does an IoU of 0 signify in terms of bounding box similarity? What about an IoU of 1?

An IoU of 0 means the two bounding boxes do not overlap at all.
An IoU of 1 means the two bounding boxes overlap perfectly, so they are identical.

(iii) CVAT Screenshots


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§2 Supervision

(c) Reflection

Unsupervised learning with autoencoders is good when the goal is to learn compact representations of data, reduce dims, or remove noise without labels. It is also helpful when we care about reconstructing the input or discovering some hidden structure in the data.

Self supervised contrastive learning is better when we want to learn representations that will later be useful for things like classification, especially when labeled data is limited. This is because contrastive methods often produce more meaningful features because they explicitly train the model to distinguish similar and dissimilar examples.

§3 Problem 3: Variational Autoencoders [20 pts]

(a) Latent Space

The latent variable z is sampled from:

$$z \sim \mathcal{N}(0, I)$$

So each latent dimension is usually assumed to be independently sampled from a normal distribution with mean 0 and variance 1.

(b) KL Divergence

(i) Explain what the terms $q_\phi(z \mid x)$ and $p_\theta(z \mid x)$ represent.

The term $q_\phi(z \mid x)$ is the encoder distributon, which is the model's approximation of the true posterior over latent variables given the data x. The term $p_\theta(z \mid x)$ is the true posterior for the generative model. It is hard compute exactly.

(ii) Name at least one mathematical property of KL-divergence that makes it a good criterion.

KL divergence is always nonnegative:

$$D_{KL}(P \,\|\, Q) \geq 0$$

iff the two distributions are the same almost everywhere. Hence it naturally measures how close the approximate posterior is to the true posterior.

(c) The ELBO

The VAE objective is

$$L_{\theta,\phi}(x) = -\mathbb{E}_{z\sim q_\phi(z|x)}[\log p_\theta(x \mid z)] + D_{KL}(q_\phi(z \mid x) \,\|\, p_\theta(z))$$

- The first term,

  $$-\mathbb{E}_{z\sim q_\phi(z|x)}[\log p_\theta(x \mid z)],$$

  is the negative expected log likelihood. It encourages the decoder to reconstruct the input x accurately from z, so minimizing this term improves reconstruction quality.

- The second term,

  $$D_{KL}(q_\phi(z \mid x) \,\|\, p_\theta(z)),$$

  is the KL regularization term so it pushes the learned distribution $q_\phi(z \mid x)$ to stay close to $p_\theta(z)$, which regularizes the latent space, makes sampling possible, and prevents the encoder from memorizing some random latent codes.

(d) Optimizing L

Starting from

$$\nabla_\phi[\text{1st term}] = \nabla_\phi \, \mathbb{E}_{z\sim q_\phi}[\log p_\theta(x \mid z)]$$

we can rephwrite this as

$$\nabla_\phi \int q_\phi(z \mid x) \log p_\theta(x \mid z) \, dz$$

$$= \int \nabla_\phi \big(q_\phi(z \mid x) \log p_\theta(x \mid z)\big) \, dz$$

Since $p_\theta(x \mid z)$ does not directly depend on ϕ:

$$= \int \log p_\theta(x \mid z) \, \nabla_\phi q_\phi(z \mid x) \, dz$$

So:

$$\nabla_\phi[\text{1st term}] = \int \log p_\theta(x \mid z)\, \nabla_\phi q_\phi(z \mid x)\, dz$$

This cant be rewritten as an expectation in the same way as the θ gradient because now the distribution we are sampling from itself depends on ϕ. So, ϕ appears inside the sampling distribution, so the gradient cannot simply pass through the expectation without additional tricks. So it makes sense why the reparameterization trick makes sense for VAEs.